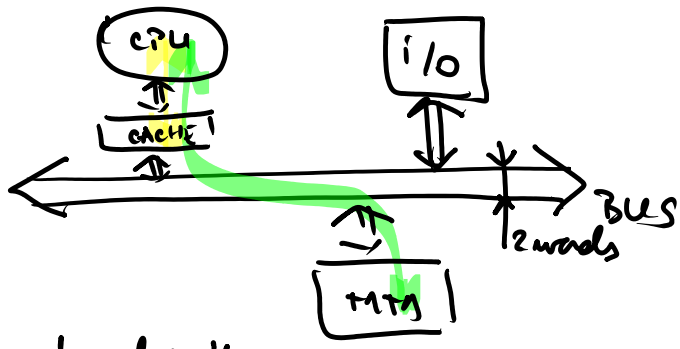
Examples

Tuesday, May 25, 2021 12:01 PM



- BUS bandwidth = 10^9 words/second
- BUS width = 2 words
- 90% of all memory accesses are found in the cache
- 1 block = 4 words, the entire block is read (allocated) in the case of a miss
- The processor sends references to the cache at a frequency of 10^8 words/sec. (reads or writes)
- 30% of all references are writes
- We assume that at any moment, 35% of all blocks have been modified (35% blocks are dirty)
- Write allocate policy for a miss

  a) The cache is Write Back

  b) The cache is Write Through

Miss Penalty in terms of MEM accesses

Total memory references = Memory references read + Memory references write

Memory references read = Cache references × Miss rate × % Reads × $\frac{\text{Memory references}}{\text{Read Miss}}$

Memory references write = Cache references × Miss rate × % Writes × $\frac{\text{Memory references}}{\text{Write Miss}}$

a) $\frac{\text{Memory references}}{\text{Write miss}} = \frac{\text{Memory references}}{\text{Read miss}} = \underbrace{\underbrace{\frac{\text{Block size}}{\text{Bus width}}}_{\text{Bus reads}}}_{\text{Allocation}} + 35\% \underbrace{\frac{\text{Block size}}{\text{Bus width}} \text{ Bus writes}}_{\text{update}}$

$= 2 \text{ Bus reads} + 35\% \times 2 \text{ Bus writes}$

$= 1.35 \times 2 \text{ Bus accesses} = 2.7 \text{ Bus}$

accesses

$$\text{Memory references}_{WB} = 10^8 \times 0.1 \times (0.7 \times 2.7 + 0.3 \times 2.7) = 10^8 \times 0.27 \text{ acc/sec.}$$

b) $$\frac{\text{Memory references}}{\text{Read miss}} = \underbrace{\frac{\text{Block size}}{\text{BUS width}}}_{\text{allocation}} \text{ BUS reads} = 2 \text{ BUS reads} = 2 \text{ BUS accesses}$$

$$\frac{\text{Memory references}}{\text{Write miss}} = \underbrace{2 \text{ BUS reads}}_{\text{allocation}} + 1 \text{ BUS Write} = 3 \text{ BUS accesses}$$

$$\frac{\text{Memory references}}{\text{Write hit}} = 1 \text{ BUS write} = 1 \text{ BUS access}$$

$$\text{Memory references}_{WT_{read}} = 10^8 \times 0.1 \times 0.7 \times 2 = \underline{0.14} \times 10^8 \text{ accesses/sec}$$

$$\text{Memory references}_{WT_{write}} = 10^8 \times (\underbrace{0.1 \times 0.3 \times 3}_{0.09} + \underbrace{0.9 \times 0.3 \times 1}_{0.27}) =$$

$$= 10^8 \times \underline{0.36} \text{ accesses/sec}$$

$$\text{Memory references}_{WT} = 10^8 \times 0.5 \text{ accesses/sec.}$$

is better

$\Rightarrow$ a) $$\text{Memory bandwidth used}_{WB} = \frac{0.27 \times 10^8}{10^9} = \frac{2.7}{10} = 2.7\%$$

b) $$\text{Memory bandwidth used}_{WT} = \frac{0.5 \times 10^8}{10^9} = \frac{5 \cdot 10^7}{10^9} = 5\%$$